Yakkala Mohan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.33
BHASHYAM JUNIOR COLLEGE	Class XII (BOARD OF INTERMEDIATE EDUCATION, ANDHRA PRADESH)	2024	96.60%
BHASHYAM HIGH SCHOOL	Class X (BOARD OF SECONDARY EDUCATION, ANDHRA PRADESH)	2022	94.16%

EXPERIENCE

Robotics Simulation Intern -- Krishaka

- Developed a ROS 2 simulation environment for an autonomous warehouse robot, integrating LiDAR, RGB/depth cameras, odometry, and TF for system analysis and validation.
- Configured ROS 2 launch files and bridge nodes to integrate sensor, transformation, and navigation data across simulation components.
- Generated 2D and 3D environment maps using RTAB-Map SLAM and evaluated localization and autonomous navigation using previously generated maps.
- Conducted simulation testing and iterative integration of perception and mapping modules, validating system behavior before deployment on physical hardware.

SCHOLASTIC ACHIEVEMENTS

- **Secured 3rd place nationally** in the **e-Yantra Robotics Competition 2025-26** at **IIT Bombay** as part of a **4-member team**, competing against **449 teams** from engineering institutes across India.

PROJECTS

Operations & Infrastructure Management System

- Built a dynamic operational dashboard to track and benchmark facility efficiency for **5,000+ residents across 23 hostels**.
- Deployed a **3-tier severity rubric**, reducing **Mean Time to Resolution (MTTR) by 40%**.
- Evaluated seasonal operational data to forecast high-risk periods, enabling **preemptive resource allocation** that prevented **20,000+ hours of aggregate downtime** without increasing the budget.

Bicycle Theft Protection System

- Identified recurring campus bicycle-theft and false-alarm problems and developed an **IoT-based security system** integrating motion sensing, Bluetooth communication, and an alarm mechanism.
- Designed a mobile interface with **Lock, Unlock, Find, and Stop Alarm** workflows for remote system control and bicycle recovery.
- Developed a motion-detection algorithm using **MPU6050 sensor data** and a movement threshold to distinguish unauthorized movement from normal activity, reducing false alarms.
- Evaluated **ESP32, GPS, LoRa, and HC-05** alternatives across communication reliability, detection capability, implementation complexity, and cost to select the final system architecture.

Multi-Robot Autonomous Warehouse Automation System

- Developed a fleet of **three autonomous robots** for warehouse pick-and-place and crate-stacking operations,

integrating perception, navigation, task allocation, and manipulation.
- Built an OpenCV-based vision pipeline for **real-time object detection and localization**, enabling autonomous decision-making from camera data.
- Designed **path-planning and multi-robot task-allocation logic** to dynamically assign operations while preventing inter-robot collisions.
- Validated system behavior in **Gazebo simulation before hardware deployment**, iterating on navigation, manipulation, and multi-robot coordination.
- Resolved payload and camera-occlusion constraints through **design iteration and mechanism redesign**, integrating a spring counterbalance and revised support structure.

Articulated Metagross Model:

Designed a **66.77 × 61 × 30.4 mm** articulated **Metagross** model in **Solid Edge**, implementing **ball-and-socket joints** to enable movable arms while ensuring printability and assembly using **FDM** additive manufacturing.

POSITIONS OF RESPONSIBILITY

- Coordinator, Informals Elan & nVision IIT Hyderabad
- Coordinator, MILAN IIT Hyderabad
- Core member, Robotix IIT Hyderabad
- Core Member, Torque IIT Hyderabad

SKILLS

- **Programming:** Python, C, MATLAB, SQL, Pandas, NumPy
- **Data & Visualization:** Power BI
- **Engineering & Simulation:** ABAQUS, Solid Edge, Fusion 360, Ansys
- **Robotics & Control:** ROS 2, Gazebo, PID Control
- **Embedded Systems & PCB Design:** ESP32, Arduino, PCB Design, LTspice
- **Manufacturing & Prototyping:** 3D Printing, Rapid Prototyping